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Experiment 1:

FIND-S Algorithm

Code:

```
# FIND-S Algorithm

# Training data
data = [
    ['Sunny', 'Warm', 'Normal',
'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High',
'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High',
'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High',
'Strong', 'Cool', 'Change', 'Yes']
]

# Initialize hypothesis
hypothesis = ['0'] * (len(data[0])
- 1)

print("Initial Hypothesis:")
print(hypothesis)

# Apply FIND-S Algorithm
for example in data:
    if example[-1] == "Yes":
        for i in
range(len(hypothesis)):
            if hypothesis[i] == '0':
                hypothesis[i] =
example[i]
            elif hypothesis[i] !=
example[i]:
                hypothesis[i] = '?'

        print("\nHypothesis after
processing", example)
        print(hypothesis)

print("\nFinal Hypothesis:")
print(hypothesis)Output:
```

Output:

```
===================== RESTART: E:\Machine Learning\exp 1.py ====================
Initial Hypothesis:
['0', '0', '0', '0', '0', '0']

Hypothesis after processing ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes']
['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']

Hypothesis after processing ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes']
['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

Hypothesis after processing ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
['Sunny', 'Warm', '?', 'Strong', '?', '?']

Final Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```